

GOVERNMENT POLYTECHNIC, AMRAVATI

Computer Department
Project 2024-25
Weekly Report on Project Development

Group No.: 4

Group Members:

1. Naman S. Bharsakale (22CM010)
2. Sarvesh S. Chaudhari (22CM016)
3. Yash S. Chaukhande (22CM019)
4. Jay A. Ganorkar (22CM030)
5. Mohammad Shehzad Mohammed Zameer (23CM206)

Guide: Mrs. C. R. Chaudhari

Week No. 1 (09/01/2025 – 15/01/2025)

Description:

The team discussed and finalized the project idea, "Smart Stick for Blind Person." Each member researched technologies related to assistive devices for visually impaired individuals.

Sr No	Name of Student	Planned Task	Implemented Task	Remarks by Guide
1	Naman S. Bharsakale	Research on smart assistive devices	Gathered details on various smart assistive devices and their working principles. 1)Link:- https://ieeexplore.ieee.org/document/10170619/ 2)Link:- https://syrmagsgs.com/assistive-devices/	Approved
2	Sarvesh S. Chaudhari	Research on obstacle detection techniques	Studied ultrasonic sensor applications for obstacle detection 1)Link:- https://www.researchgate.net/publication/337842014 2)Link:- https://ieeexplore.ieee.org/document/10303550	Suggested additional sensor study
3	Yash Chaukhande	Research on fire detection methods	Collected data on flame sensors used in fire detection 1)Link:- https://www.winsen-sensor.com/knowledge 2)Link:- https://www.elprocus.com	Approved

4	Jay Ganorkar	Study of emergency notification systems	Studied GPS and Pushbullet API for emergency alerts 1)Link:- https://docs.pushbullet.com/ 2)Link:- https://www.researchgate.net/publication/388962706	Suggested real-time testing
5	Mohammad Shehzad Mohammed Zameer	Component selection and feasibility study	Identified necessary hardware components like ESP32, sensors, and power modules 1)Link:- https://www.nabto.com/guide-to-iot-esp-32/ 2)Link:- https://www.researchgate.net	Need to check availability

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Week No. 2 (16/01/2025 – 22/01/2025)

Description:

Selected components were analyzed for feasibility and working. Component procurement was initiated.

Sr No	Name of Student	Planned Task	Implemented Task	Remarks by Guide
1	Naman S. Bharsakale	Testing ultrasonic sensor	Successfully detected obstacles within 1.5 meters. 1)Link:- https://www.sciencedirect.com 2)Link:- https://mycardoeswhat.org	Approved
2	Sarvesh S. Chaudhari	Testing flame sensor	Successfully detected fire at different distances 1)Link:- https://forum.arduino.cc/t/how-find-fire-from-long-distance/17973 2)Link:- https://docs.espressif.com/projects/esp-idf/en/stable/esp32/get-started/establish-serial-connection.html	Check sensor accuracy
3	Yash Chaukhande	Configuring ESP32	Established basic connectivity with ESP32 1)Link:- https://docs.espressif.com/projects/esp-idf/en/stable/esp32/get-started/establish-serial-connection.html 2)Link:- https://lastminuteengineers.com/getting-started-with-esp32/	Approved
4	Jay Ganorkar	Testing GPS module	GPS coordinates retrieved successfully 1)Link:- https://www.techtarget.com/whatis/definition/GPS-coordinates 2)Link:- https://www.circuitdigest.com/microcontroller-projects/simple-gps-tracker-using-esp32-visualize-data-on-map	Need real-world testing
5	Mohammad Shehzad Mohammed Zameer	Checking power supply	Identified optimal battery and regulator for power stability 1)Link:- https://www.sciencedirect.com/science/article/abs/pii/S2352152X22006260	Approved

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Week No. 3 (23/01/2025 – 29/01/2025)

Description:

Basic circuit setup and initial code implementation.

Sr No	Name of Student	Planned Task	Implemented Task	Remarks by Guide
1	Naman S. Bharsakale	Connecting sensors to ESP32	Successfully connected ultrasonic and flame sensors 1)Link:- https://www.linkedin.com/pulse/how-interface-ultrasonic-sensor-hc-sr04-esp32-using-parvathaneni-p10tc 2)Link:- https://projecthub.arduino.cc/SURYATEJA/arduino-modules-flame-sensor-e48e97	Approved
2	Sarvesh S. Chaudhari	Writing initial sensor code	Basic sensor readings obtained 1)Link:- https://youtube.com/shorts/2d2p1-7KgJg?si=OrpBkm2SpoloBbXW	Optimize code efficiency
3	Yash Chaukhande	Testing vibration motor	Motor triggered successfully upon obstacle detection 1)Link:- https://ijrpr.com/uploads/V5ISSUE7/IJRPR31406.pdf 2)Link:- https://www.sciencebuddies.org/science-fair-projects/project-ideas/Elec_p107/electricity-electronics/obstacle-detecting-walking-stick	Increase vibration intensity
4	Jay Ganorkar	Implementing emergency SMS notification	Pushbullet API tested with sample messages 1)Link:- https://www.pushbullet.com/api 2)Link:- https://docs.pushbullet.com/v4/	Approved
5	Mohammad Shehzad Mohammed Zameer	Powering up the circuit	Ensured stable power flow to all components 1)Link:- https://techexplorations.com/guides/esp32/begin/power/	Approved

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Week No. 4 (30/01/2025 – 05/02/2025)

Description:

Refinement of sensor accuracy and debugging issues in the prototype.

Sr No	Name of Student	Planned Task	Implemented Task	Remarks by Guide
1	Naman S. Bharsakale	Optimize obstacle detection	Improved accuracy using better threshold values 1)Link:- https://help.salesforce.com/s/articleView?id=release-notes.rn_bi_edd_model_threshold_deploy.htm&release=222&type=5 2)Link:- https://www.evidentlyai.com/classification-metrics/classification-threshold	Approved
2	Sarvesh S. Chaudhari	Improve flame sensor accuracy	Adjusted sensitivity for better fire detection 1)Link:- https://youtu.be/vzKjqQZ65-Y?si=zE77cZgcdEyTddTn	Check for false alarms
3	Yash Chaukhande	Debugging ESP32 connectivity issues	Resolved minor connectivity errors 1)Link:- https://randomnerdtutorials.com/esp32-troubleshooting-guide/ 2)Link:- http://docs.espressif.com/projects/esptool/en/latest/esp32/troubleshooting.html	Approved
4	Jay Ganorkar	Testing emergency alert reliability	Pushbullet API tested with multiple recipients 1)Link:- https://docs.pushbullet.com/	Optimize response time
5	Mohammad Shehzad Mohammed Zameer	Improving circuit stability	Added resistors to reduce voltage fluctuations 1)Link:- https://electronics.stackexchange.com/questions/127525/reducing-voltage-with-resistors 2)Link:- https://forum.arduino.cc/t/what-is-the-correct-way-to-drop-voltage-in-circuit/1264735/2	Approved

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Week No. 5 (06/02/2025 – 12/02/2025)

Description:

Integration of all sensors and modules, checking real-time performance.

Components:

1. **Ultrasonic Sensor:** Measures distance by emitting ultrasonic waves and calculating the time it takes for the echo to return.
2. **Flame Sensor:** Detects the presence of fire or flame using infrared light sensing.
3. **GPS Module:** Receives signals from satellites to determine precise geographic location coordinates.
4. **Vibrating Motor:** Produces vibration feedback when powered, commonly used for alerts in devices.
5. **Rechargeable Battery:** Stores electrical energy and can be recharged multiple times for repeated use.
6. **ESP32:** A low-cost, low-power microcontroller with built-in Wi-Fi and Bluetooth capabilities.
7. **Push Button:** A simple switch mechanism used to control circuits by making or breaking a connection when pressed.

Sr No	Name of Student	Planned Task	Implemented Task	Remarks by Guide
1	Naman S. Bharsakale	Combine sensor data	Successfully integrated obstacle and fire detection 1)Link:- https://www.researchgate.net/publication/353795465_Smart_Blind_Stick_Using_Ultrasonic_Sensor	Approved
2	Sarvesh S. Chaudhari	Improve code structure	Optimized sensor readings and response time 1)Link:- https://projecthub.arduino.cc/SURYATEJA/arduino-modules-flame-sensor-e48e97	Approved
3	Yash Chaukhande	Test vibration motor response	Vibration motor triggered correctly on detection 1)Link:- https://gesrepair.com/common-causes-of-motor-vibration/ 2)Link:- https://teamcorporation.com/2023/03/27/vibration-testing-methods/	Increase motor duration

Sr No	Name of Student	Planned Task	Implemented Task	Remarks by Guide
4	Jay Ganorkar	Improve GPS module accuracy	Location tracking refined for emergency alerts 1)Link:- https://nerdschalk.com/improve-location-accuracy-android/ 2)Link:- https://en.wikipedia.org/wiki/Differential_GPS	Approved
5	Mohammad Shehzad Mohammed Zameer	Monitor power consumption	Identified excessive power draw and optimized circuit 1)Link:- https://www.sciencedirect.com/topics/computer-science/power-consumption	Approved

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Week No. 6 (13/02/2025 – 19/02/2025)

Description:

Field testing of the prototype with real-time obstacle and fire detection.

Sr No	Name of Student	Planned Task	Implemented Task	Remarks by Guide
1	Naman S. Bharsakale	Test stick in real-world conditions	Successfully detected obstacles outdoors 1)Link:- https://www.mdpi.com/2227-7080/10/3/54	Approved
2	Sarvesh S. Chaudhari	Analyze fire sensor response time	Fire detection improved with threshold tuning 1)Link:- https://www.sciencedirect.com/science/article/abs/pii/S0379711224000018	Approved
3	Yash Chaukhande	Vibration motor field test	Motor response tested on different surfaces 1)Link:- https://www.elementdefense.com/blog/vibration-testing-applications 2)Link:- https://www.bksv.com/media/doc/bn2352.pdf	Approved
4	Jay Ganorkar	Test emergency alert in different locations	GPS tracking works accurately within 5m 1)Link:- https://www.fema.gov/emergency-managers/practitioners/integrated-public-alert-warning-system/public/wireless-emergency-alerts/geographic-accuracy-wea	Approved
5	Mohammad Shehzad Mohammed Zameer	Check battery backup in real use	Battery lasts 6 hours on full charge 1)Link:- https://www.averna.com/mastering-battery-testing-everything-you-need-to-know	Consider power optimization

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Week No. 7 (20/02/2025 – 26/02/2025)

Description:

Improving battery efficiency and refining emergency alert system.

Sr No	Name of Student	Planned Task	Implemented Task	Remarks by Guide
1	Naman S. Bharsakale	Reduce power consumption	Lowered ESP32 power usage 1)Link:- https://www.espboards.dev/blog/esp32-power-optimisation/ 2)Link:- http://diyi0t.com/reduce-the-esp32-power-consumption/	Approved
2	Sarvesh S. Chaudhari	Optimize fire detection power usage	Adjusted sensor sleep mode for efficiency 1)Link:- https://www.sciencedirect.com/science/article/pii/S2665917424003805	Approved
3	Yash Chaukhande	Improve vibration motor efficiency	Reduced power draw with PWM 1)Link:- https://www.precisionmicrodrives.com/ab-012	Approved
4	Jay Ganorkar	Enhance emergency SMS speed	Pushbullet messages sent within 3 seconds 1)Link:- https://help.pushbullet.com/articles/why-are-my-text-messages-stuck-pending/	Approved
5	Mohammad Shehzad Mohammed Zameer	Test alternative power sources	Explored solar charging option 1)Link:- https://randomnerdtutorials.com/power-esp32-esp8266-solar-panels-battery-level-monitoring/	Approved

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Week No. 8 (27/02/2025 – 05/03/2025)

Description:

Preparing a proper stick enclosure and finalizing design.

Sr No	Name of Student	Planned Task	Implemented Task	Remarks by Guide
1	Naman S. Bharsakale	Design enclosure for sensors	Created a 3D-printed case 1)Link:- https://www.allaboutcircuits.com/industry-articles/six-steps-for-designing-a-custom-3d-printed-electronics-enclosure/	Approved
2	Sarvesh S. Chaudhari	Mount flame sensor securely	Placed sensor in optimal position 1)Link:- https://www.hvac.com/expert-advice/furnace-flame-sensors/	Approved
3	Yash Chaukhande	Secure vibration motor	Fixed motor with shock-absorbing material 1)Link:- https://us.misumiec.com/vona2/detail/221004938469/?srsltid=AfmBOorOKTfEzLg16mQuiNtT2CRHlOfPkrmG3NQCMjaoXx4SY5Pdzi4	Approved
4	Jay Ganorkar	Attach GPS module properly	Placed GPS externally for better signal 1)Link:- https://support.avenzamaps.com/hc/en-us/articles/4961419399956-Connecting-to-an-External-GPS-Device	Approved
5	Mohammad Shehzad Mohammed Zameer	Ensure all connections are durable	Used heat shrink tubes for safety 1)Link:- https://axis-india.com/heat-shrink-tubing-the-analysis-you-need/	Approved

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Week No. 9 (06/03/2025 – 12/03/2025)

Description:

Testing and debugging final prototype for reliability.

Sr No	Name of Student	Planned Task	Implemented Task	Remarks by Guide
1	Naman S. Bharsakale	Test durability of stick	Conducted stress tests 1)Link:- https://blog.qima.com/electronics/overview-test-methods-for-electronic-components 2)Link:- https://resources.altium.com/p/overview-electrical-stress-test-methods-pcbas	Approved
2	Sarvesh S. Chaudhari	Debug fire sensor for false alarms	Improved accuracy with software filtering 1)Link:- https://www.mdpi.com/1424-8220/22/9/3307	Approved
3	Yash Chaukhande	Verify vibration motor consistency	Ensured proper motor response in all cases 1)Link:- https://forum.arduino.cc/t/arduino-and-multiple-vibration-motors/118441	Approved
4	Jay Ganorkar	Ensure emergency alert reliability	Multiple successful tests with different networks 1)Link:- https://www.fema.gov/emergency-managers/practitioners/integrated-public-alert-warning-system/public/emergency-alert-system	Approved
5	Mohammad Shehzad Mohammed Zameer	Check full system battery backup	Lasts 8 hours after optimizations 1)Link:- https://embeddedartistry.com/blog/2020/10/26/optimizing-power-consumption-in-battery-powered-devices/	Approved

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Week No. 10 (13/03/2025 – 19/03/2025)

Description:

User testing and collecting feedback from visually impaired individuals.

Sr No	Name of Student	Planned Task	Implemented Task	Remarks by Guide
1	Naman S. Bharsakale	Test with visually impaired users	Gathered valuable feedback 1)Link:- https://www.researchgate.net/publication/356441226_A_systematic_literature_review_on_the_usability_of_mobile_applications_for_visually_impaired_users 2)Link:- https://www.researchgate.net/publication/378254054	Approved
2	Sarvesh S. Chaudhari	Observe real-world usability	Users found obstacle detection helpful 1)Link:- https://www.sciencedirect.com/science/article/pii/S2090447923002769 2)Link:- https://www.testdevlab.com/blog/usability-testing-in-software-development	Approved
3	Yash Chaukhande	Record user response to vibration alerts	Users adapted quickly to alerts 1)Link:- https://www.researchgate.net/publication/345370472_Assistive_Mobile_Application_for_Visually_Impaired_People	Approved
4	Jay Ganorkar	Evaluate emergency alert effectiveness	Emergency messages sent successfully 1)Link:- https://youtu.be/E-5vazlDznc?si=sLQ_3DwLO3ig4AYL	Approved
5	Mohammad Shehzad Mohammed Zameer	Note user battery concerns	Users requested longer battery life	Approved

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Week No. 11 (20/03/2025 – 26/03/2025)

Description:

Making final modifications and preparing documentation.

Sr No	Name of Student	Planned Task	Implemented Task	Remarks by Guide
1	Naman S. Bharsakale	Adjust system based on user feedback	Made necessary changes 1)Link:- https://www.linkedin.com/advice/0/how-do-you-incorporate-user-feedback-system-improvement	Approved
2	Sarvesh S. Chaudhari	Improve fire detection responsiveness	Enhanced response speed 1)Link:- https://www.mdpi.com/2076-3417/13/13/7845	Approved
3	Yash Chaukhande	Ensure motor durability	Confirmed longevity of motor 1)Link:- https://www.divaportal.org/smash/get/diva2%3A833367/FULLTEXT01.pdf	Approved
4	Jay Ganorkar	Finalize GPS and alert system	Improved alert reliability 1)Link:- https://youtu.be/vi_cluxDpcA?si=OBKn3uhSSw6W_HoE	Approved
5	Mohammad Shehzad Mohammed Zameer	Complete documentation	Project report drafted 1)Link:- https://www.project.co/project-report/	Approved

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Week No. 12 (27/03/2025 – 02/04/2025)

Description:

Finalizing project report, testing the smart stick one last time, preparing for the final presentation, and submitting all required documents.

Sr No	Name of Student	Planned Task	Implemented Task	Remarks by Guide
1	Naman S. Bharsakale	Review final prototype performance	Last round of field testing, confirmed all features work	Approved
2	Sarvesh S. Chaudhari	Fix minor software bugs	Optimized final code and removed unnecessary delays	Approved
3	Yash Chaukhande	Ensure all hardware connections are stable	Checked soldering and wiring for durability	Approved
4	Jay Ganorkar	Finalize emergency alert system with Pushbullet	Verified alert speed and reliability under various conditions	Approved
5	Mohammad Shehzad Mohammed Zameer	Check battery performance in long-term use	Ensured battery lasts a minimum of 8 hours	Approved
6	All Members	Compile final project report	Report structured with detailed explanation, images, and code snippets	Approved
7	All Members	Prepare PowerPoint presentation	Completed with diagrams, graphs, and explanation of each module	Approved

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